

STUDENT INFORMATION SYSTEM – PROJECT DOCUMENTATION

1. PROJECT OVERVIEW

The Student Information System is a simple Java console-based application designed to store, manage, and display student details such as Student ID, Name, Age, Grade, and Contact Information.

2. OBJECTIVES

1. The objective of this project is to design a Student class using object-oriented principles to store essential student details.
2. The system ensures data accuracy by applying validation for age and grade values.
3. A menu-driven console application is developed using the Scanner class for smooth user interaction.
4. The project supports CRUD operations and searching of student records by ID or name. Student data is managed dynamically using arrays or ArrayList during runtime.
5. Proper output formatting and basic error handling are implemented to improve usability.

3. SETUP & INSTALLATION INSTRUCTIONS

1. To run the Student Information System project, ensure that Java Development Kit (JDK) 8 or higher is installed on the system. The JDK can be downloaded from the OpenJDK website. After installation, verify Java installation by running `java -version` in the command prompt.
2. Next, install any Java-supported Integrated Development Environment (IDE) such as Eclipse, a new Java project and name it appropriately (for example, *StudentInformationSystem*).
3. Inside the project, create a package named `INTERNSHIP`. Add the required Java class files: `Student.java`, `StudentManager.java`, `StudentInformationSystem.java`, and `ValidationUtils.java`. write respective source code into these files and save them.
4. Compile the project to check for errors. Run the `StudentInformationSystem.java` file, which contains the `main()` method. The menu-driven Student Information System will appear in the console, and the application will be ready for use.

Project Structure:

- `Student.java` - Student class definition

- StudentInformationSystem.java - Main program with menu
- StudentManager.java - Business logic class
- ValidationUtils.java - Input validation methods

4. Code Structure Explanation

STUDENT CLASS

The Student class represents a student entity. It contains private attributes for student ID, name, age, grade, and contact. A constructor is used to initialize these values. Getter and setter methods allow controlled access to the data. The toString() method is overridden to display student details in a formatted and readable way using tab spacing.

```
package INTERNSHIP;

public class Student {

    private String studentId;
    private String name;
    private int age;
    private double grade;
    private String contact;

    public Student(String studentId,String name,int age,double grade,String contact) {
        this.studentId=studentId;
        this.name=name;
        this.age=age;
        this.grade=grade;
        this.contact=contact;
    }

    public String getStudentId() {
        return studentId;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}
```

```

    public int getAge() {
        return age;
    }

    public void setAge(int age) {
        this.age = age;
    }

    public double getGrade() {
        return grade;
    }

    public void setGrade(double grade) {
        this.grade = grade;
    }

    public String getContact() {
        return contact;
    }

    public void setContact(String contact) {
        this.contact = contact;
    }

    @Override
    public String toString() {
        return studentId + "\t\t"
            + name + "\t\t"
            + age + "\t\t"
            + grade + "\t\t"
            + contact;
    }
}

```

STUDENTINFORMATIONSYSTEM CLASS

This is the main driver class of the application. It uses the Scanner class to take user input and displays a menu-driven interface. Based on the user's choice, it calls appropriate methods from the

StudentManager class to perform operations like add, view, search, update, and delete. Input validation loops are used to ensure correct age and grade values.

```
package INTERNSHIP;

import java.util.Scanner;

public class StudentInformationSystem {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        StudentManager manager = new StudentManager();
        int choice;

        do {
            System.out.println("\n=== STUDENT INFORMATION SYSTEM ===");
            System.out.println("1. Add New Student");
            System.out.println("2. View All Students");
            System.out.println("3. Search Student");
            System.out.println("4. Update Student");
            System.out.println("5. Delete Student");
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");

            while (!sc.hasNextInt()) {
                System.out.print("Invalid choice! Enter again: ");
                sc.next();
            }
            choice = sc.nextInt();
            sc.nextLine();

            switch (choice) {

                case 1:
                    System.out.println("\n=== ADD NEW STUDENT ===");

                    System.out.print("Enter Student ID: ");
                    String id = sc.nextLine();

                    System.out.print("Enter Name: ");
                    String name = sc.nextLine();

                    System.out.print("Enter Age: ");
                    int age = sc.nextInt();
                    while (!ValidationUtils.validateAge(age)) {
                        System.out.print("Invalid age! Enter again: ");
                        age = sc.nextInt();
                    }

                    System.out.print("Enter Grade: ");
                    double grade = sc.nextDouble();
                    sc.nextLine();
```

```

while (!ValidationUtils.validateGrade(grade)) {
    System.out.print("Invalid grade! Enter 0-100: ");
    grade = sc.nextDouble();
    sc.nextLine();
}

System.out.print("Enter Contact: ");
String contact = sc.nextLine();

manager.addStudent(new Student(id, name, age, grade, contact));
System.out.println("Student added successfully!");
break;

case 2:

    manager.viewStudents();
    break;

case 3:
    System.out.println("\n=== SEARCH STUDENT ===");
    System.out.print("Enter Student ID or Name: ");
    String key = sc.nextLine();
    Student s = manager.searchStudent(key);

    if (s != null) {
        System.out.println("\nStudent Found:");
        System.out.println(s);
    } else {
        System.out.println("Student not found!");
    }
    break;

case 4:
    System.out.println("\n=== UPDATE STUDENT ===");
    System.out.print("Enter Student ID: ");
    String uid = sc.nextLine();
    Student st = manager.searchStudent(uid);

    if (st != null) {

        System.out.print("Enter New Name: ");
        st.setName(sc.nextLine());

        System.out.print("Enter New Age: ");
        int newAge = sc.nextInt();
        while (!ValidationUtils.validateAge(newAge)) {
            System.out.print("Invalid age! Enter again: ");
            newAge = sc.nextInt();
        }
        st.setAge(newAge);

        System.out.print("Enter New Grade: ");
        double newGrade = sc.nextDouble();

```

```

        sc.nextLine();
        while (!ValidationUtils.validateGrade(newGrade)) {
            System.out.print("Invalid grade! Enter 0-100: ");
            newGrade = sc.nextDouble();
            sc.nextLine();
        }
        st.setGrade(newGrade);

        System.out.print("Enter New Contact: ");
        st.setContact(sc.nextLine());

        System.out.println("Student updated successfully!");

    } else {
        System.out.println("Student not found!");
    }
    break;

case 5:
    System.out.println("\n=== DELETE STUDENT ===");
    System.out.print("Enter Student ID to delete: ");
    String did = sc.nextLine();

    if (manager.deleteStudent(did)) {
        System.out.println("Student deleted successfully!");
    } else {
        System.out.println("Student not found!");
    }
    break;

case 6:
    System.out.println("Exiting program...");
    break;

default:
    System.out.println("Invalid choice!");
}

} while (choice != 6);

sc.close();
}
}

```

STUDENTMANAGER CLASS

The StudentManager class contains the business logic of the application. It maintains an ArrayList<Student> to store student objects. Methods are provided to add new students, display all

students, search students by ID or name, and delete students. This class ensures proper separation of logic from user interaction.

```
package INTERNSHIP;

import java.util.ArrayList;

public class StudentManager {

    private ArrayList<Student> students = new ArrayList<>();

    public void addStudent(Student s) {
        students.add(s);
    }

    public void viewStudents() {
        if (students.isEmpty()) {
            System.out.println("No student records found.");
            return;
        }

        System.out.println("=== ALL STUDENTS ===");
        System.out.println("Student ID\tName\tAge\tGrade\tContact");
        System.out.println("-----");

        for (Student s : students) {
            System.out.println(s);
        }
    }

    public Student searchStudent(String keyword) {
        for (Student s : students) {
            if (s.getStudentId().equalsIgnoreCase(keyword) ||
                s.getName().equalsIgnoreCase(keyword)) {
                return s;
            }
        }
        return null;
    }

    public boolean deleteStudent(String studentId) {
        Student s = searchStudent(studentId);
        if (s != null) {
            students.remove(s);
            return true;
        }
        return false;
    }
}
```

VALIDATIONUTILS CLASS

The ValidationUtils class handles input validation. It contains static methods to validate age and grade values. This keeps validation logic separate and reusable, improving code clarity and maintainability.

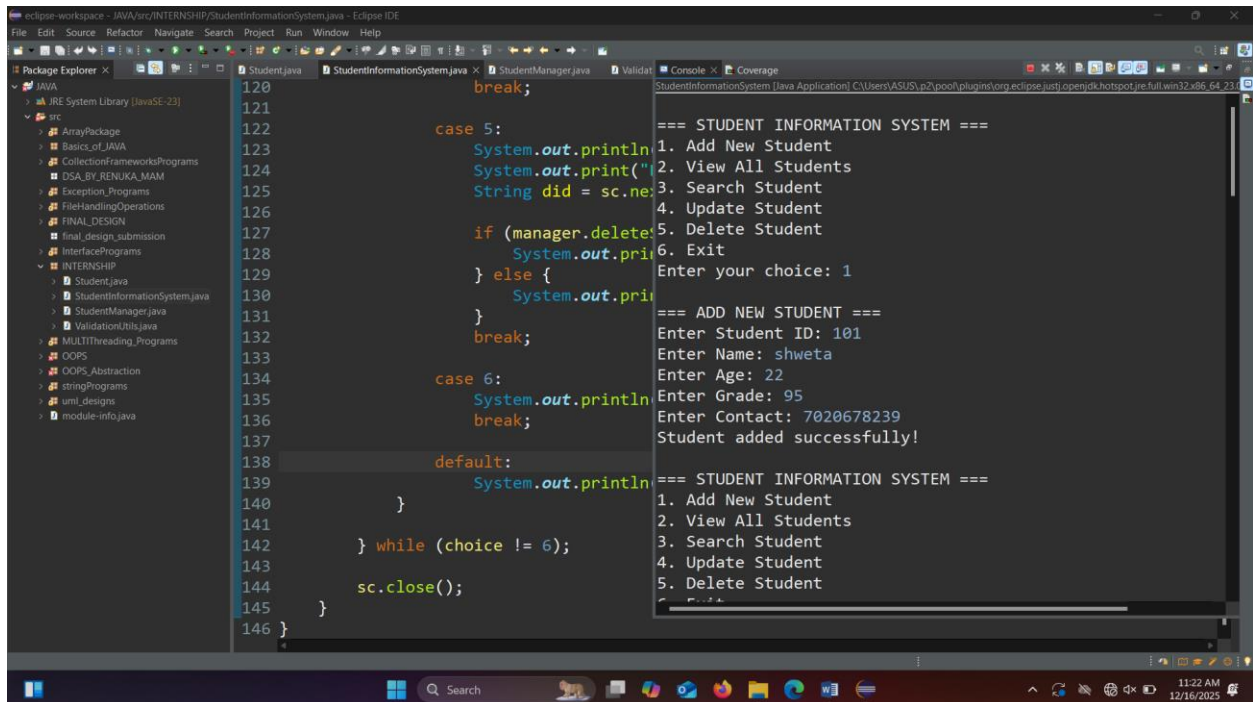
```
package INTERNSHIP;

public class ValidationUtils {
    public static boolean validateAge(int age) {
        return age > 0;
    }

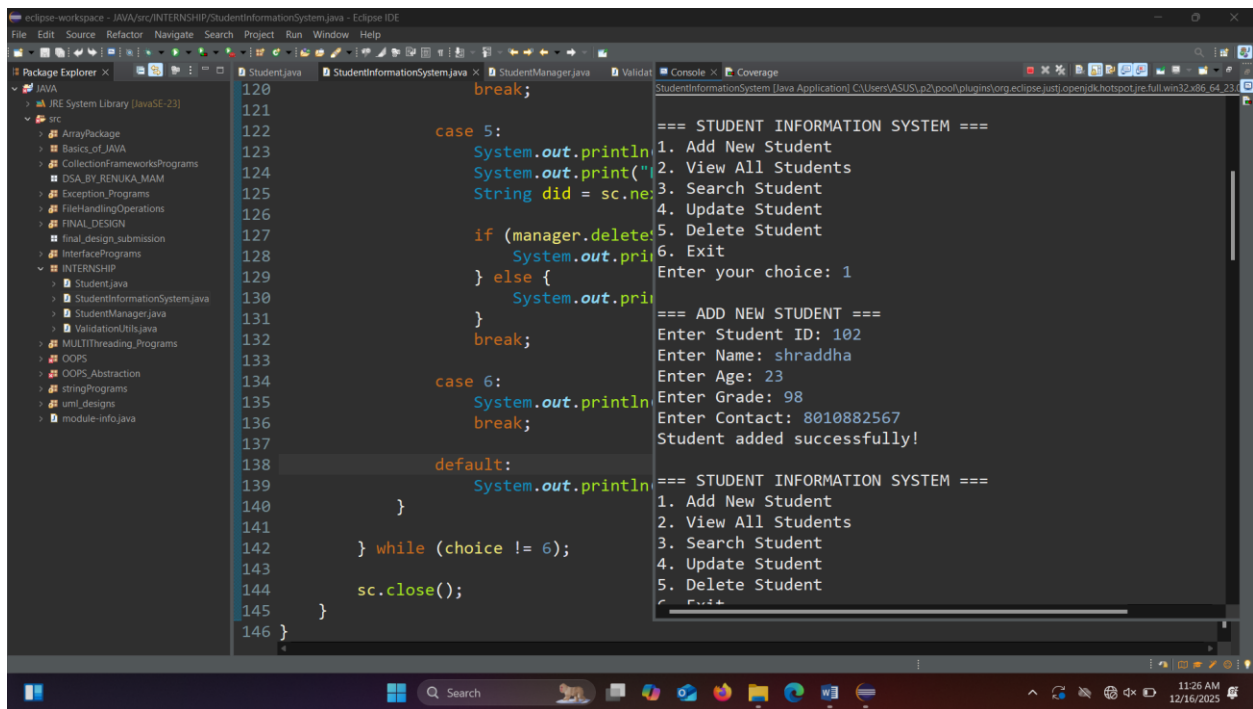
    public static boolean validateGrade(double grade) {
        return grade >= 0 && grade <= 100;
    }
}
```

5. SCREENSHOTS OF WORKING APPLICATION

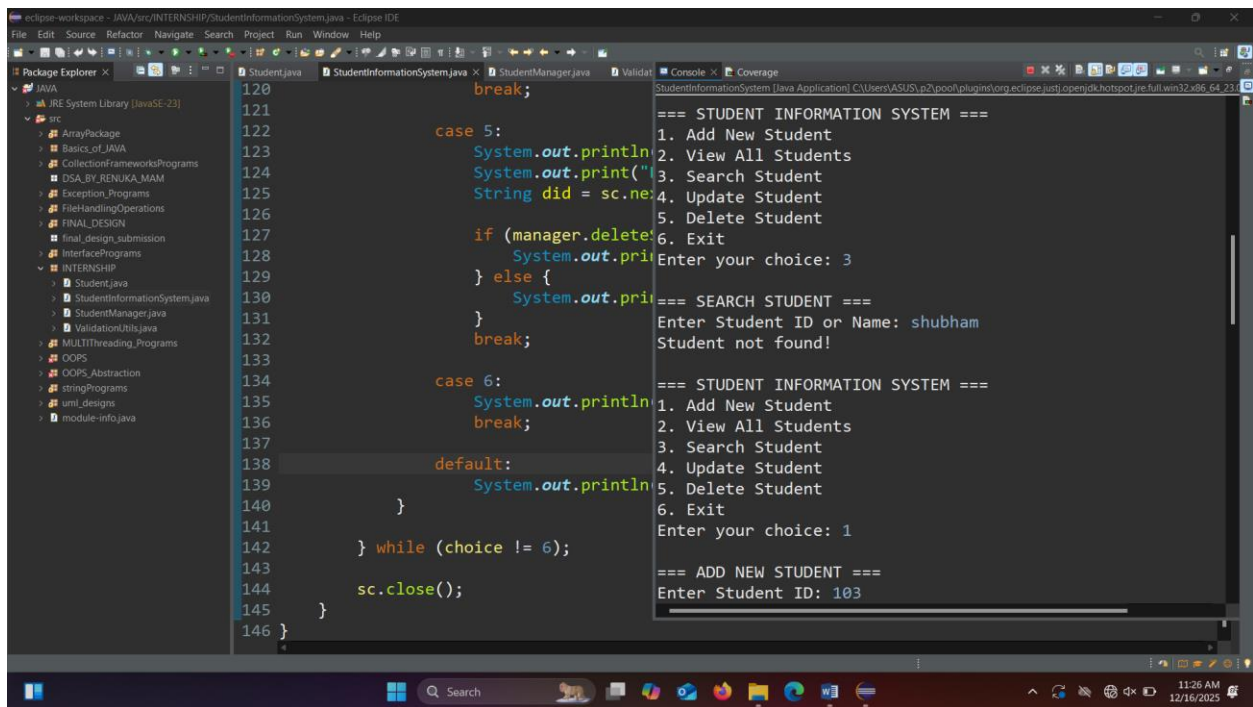
First Add New Student Operation

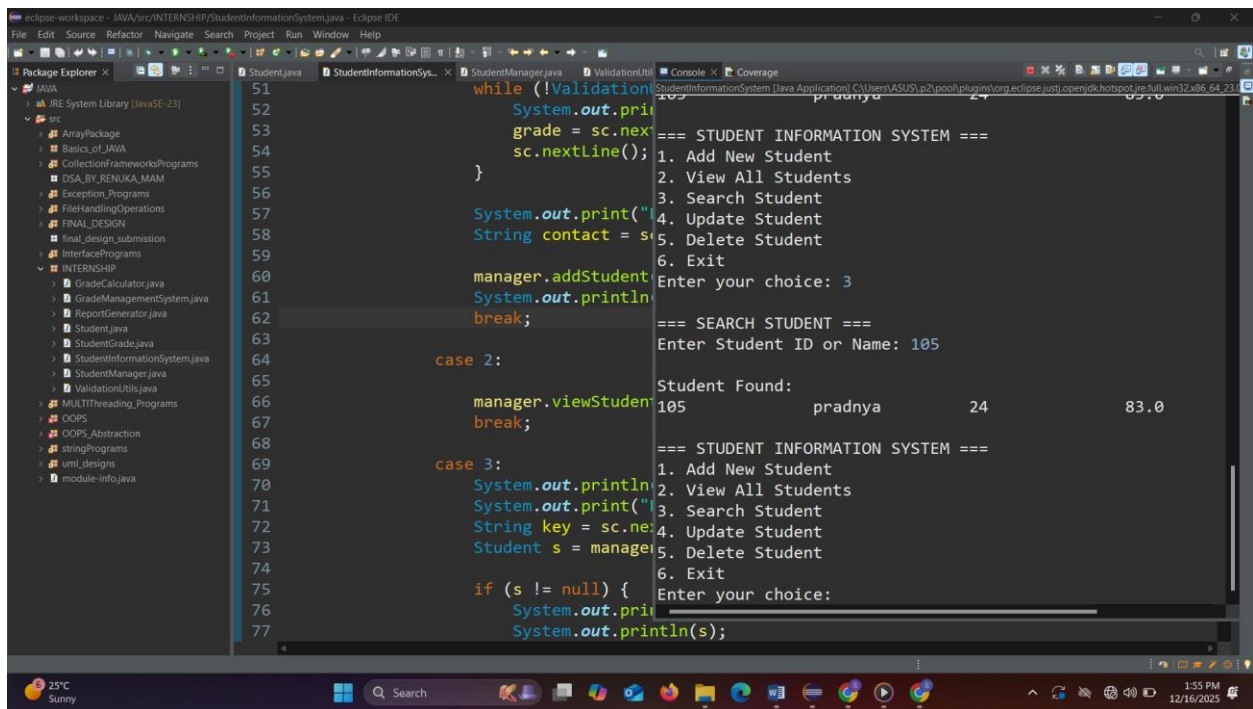


Add Student

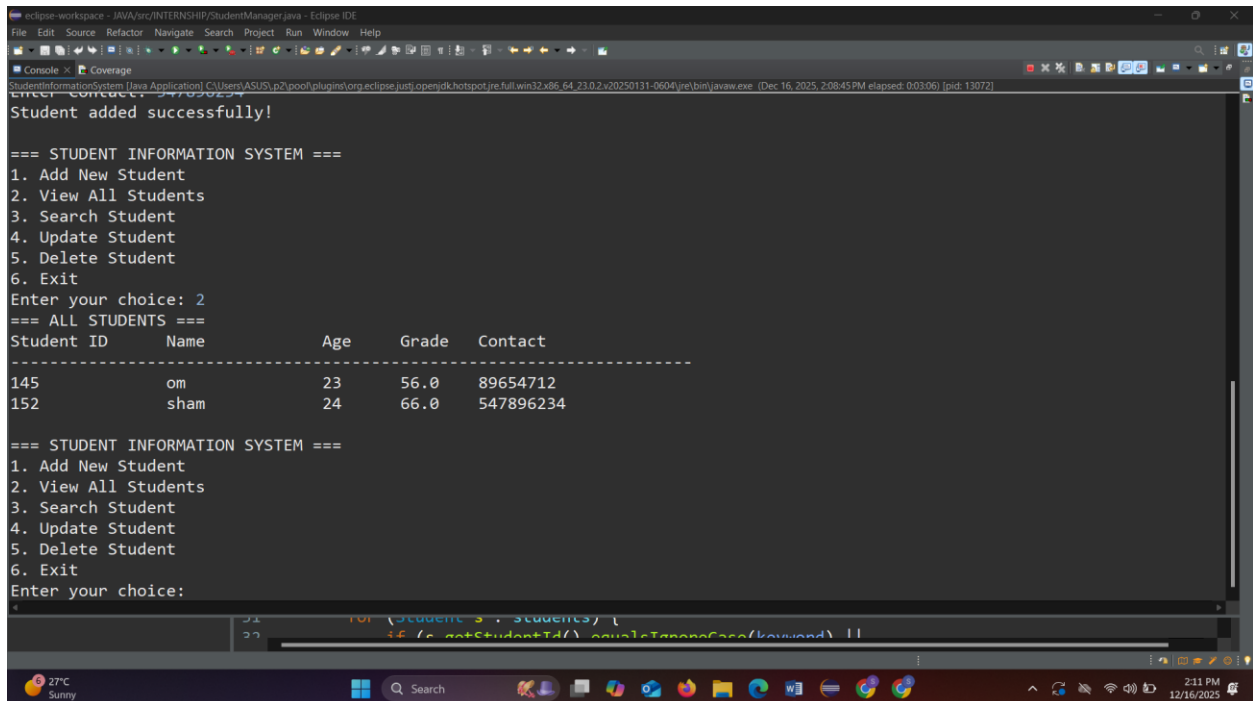


Search Student

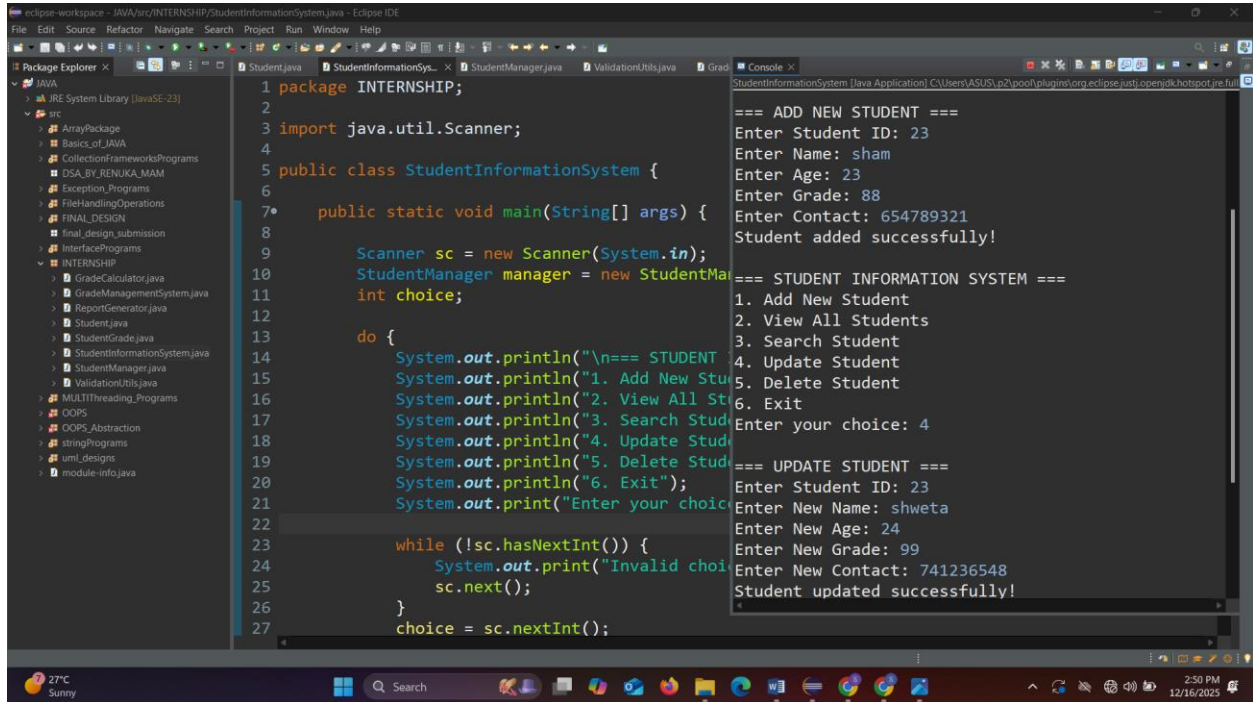




View All Student



Update Student Information



The screenshot shows the Eclipse IDE with the 'StudentInformationSystem.java' file open. The code implements a menu-driven system for updating student information. The console output shows the user selecting option 4, entering a student ID of 23, a new name 'shweta', a new age of 24, and a new grade of 99, resulting in a successful update.

```
1 package INTERNSHIP;
2
3 import java.util.Scanner;
4
5 public class StudentInformationSystem {
6
7     public static void main(String[] args) {
8
9         Scanner sc = new Scanner(System.in);
10        StudentManager manager = new StudentManager();
11        int choice;
12
13        do {
14            System.out.println("\n=== STUDENT INFORMATION SYSTEM ===");
15            System.out.println("1. Add New Student");
16            System.out.println("2. View All Students");
17            System.out.println("3. Search Student");
18            System.out.println("4. Update Student");
19            System.out.println("5. Delete Student");
20            System.out.println("6. Exit");
21            System.out.print("Enter your choice: ");
22
23            while (!sc.hasNextInt()) {
24                System.out.print("Invalid choice. Enter a valid integer: ");
25                sc.next();
26            }
27            choice = sc.nextInt();
28
29            switch (choice) {
30                case 1:
31                    manager.addStudent();
32                    break;
33                case 2:
34                    manager.viewStudents();
35                    break;
36                case 3:
37                    manager.searchStudent();
38                    break;
39                case 4:
40                    manager.updateStudent();
41                    break;
42                case 5:
43                    manager.deleteStudent();
44                    break;
45                case 6:
46                    System.out.println("Exiting the system. Goodbye!");
47                    return;
48            }
49        } while (true);
50    }
51}
```

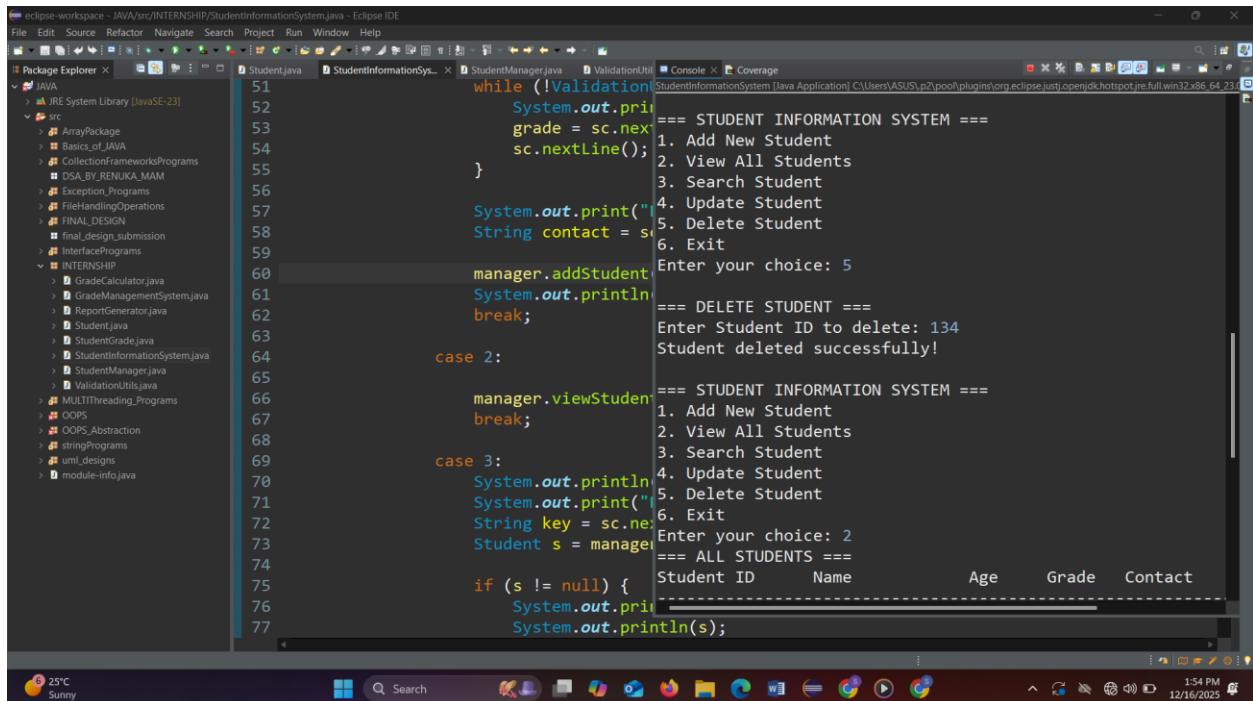
Console Output:

```
=== ADD NEW STUDENT ===
Enter Student ID: 23
Enter Name: sham
Enter Age: 23
Enter Grade: 88
Enter Contact: 654789321
Student added successfully!

=== STUDENT INFORMATION SYSTEM ===
1. Add New Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 4

=== UPDATE STUDENT ===
Enter Student ID: 23
Enter New Name: shweta
Enter New Age: 24
Enter New Grade: 99
Enter New Contact: 741236548
Student updated successfully!
```

Delete Student



The screenshot shows the Eclipse IDE with the 'StudentInformationSystem.java' file open. The code implements a menu-driven system for deleting a student. The console output shows the user selecting option 5, entering a student ID of 134, and successfully deleting the student.

```
51 while (!ValidationUtil.isValidInteger(contact)) {
52     System.out.print("Invalid contact number. Enter a valid integer: ");
53     grade = sc.nextInt();
54     sc.nextLine();
55 }
56
57 System.out.print("Enter contact number: ");
58 String contact = sc.nextLine();
59
60 manager.addStudent();
61 System.out.println("Student added successfully!");
62 break;
63
64 case 2:
65     manager.viewStudents();
66     break;
67
68 case 3:
69     System.out.println("Searching for student...");
70     System.out.print("Enter student ID to search: ");
71     String key = sc.nextLine();
72     Student s = manager.searchStudent(key);
73
74     if (s != null) {
75         System.out.println("Student found:");
76         System.out.println(s);
77     } else {
78         System.out.println("Student not found.");
79     }
80 } while (true);
81 }
```

Console Output:

```
=== STUDENT INFORMATION SYSTEM ===
1. Add New Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 5

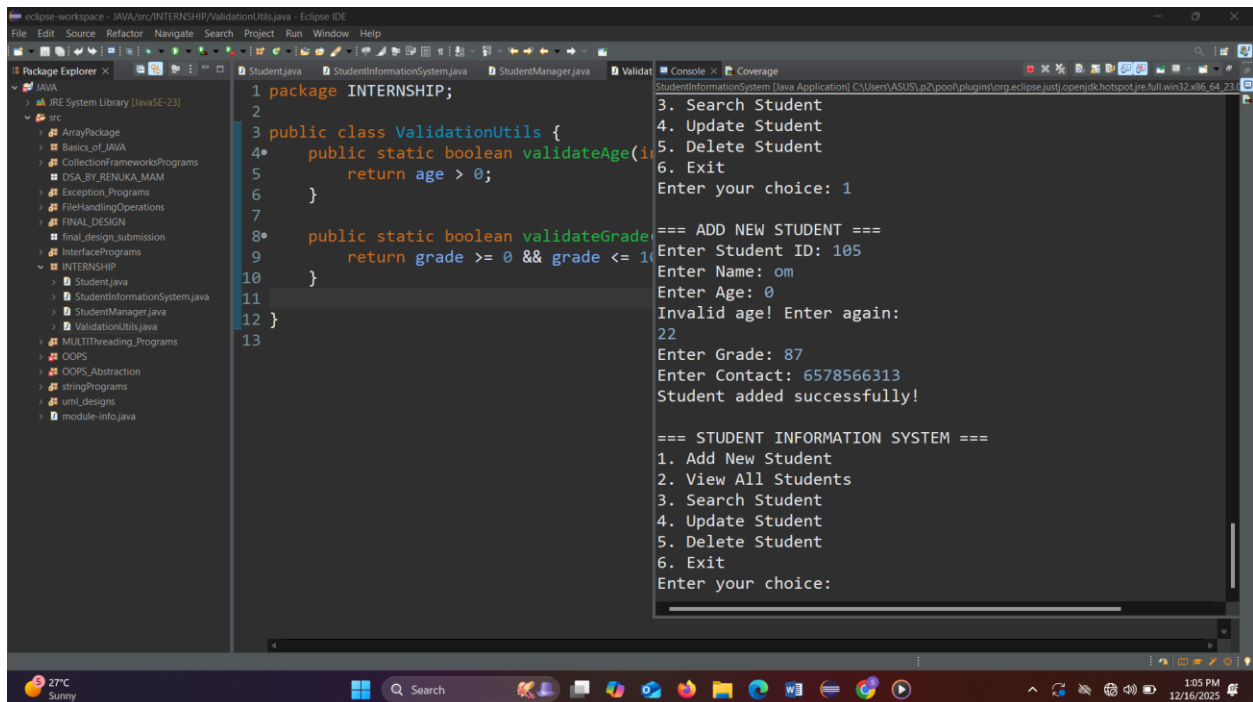
=== DELETE STUDENT ===
Enter Student ID to delete: 134
Student deleted successfully!

=== STUDENT INFORMATION SYSTEM ===
1. Add New Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 2

=== ALL STUDENTS ===
Student ID      Name      Age      Grade      Contact
-----
101      sham      23      88      654789321
102      sham      23      88      654789321
103      sham      23      88      654789321
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6. TESTING EVIDENCE

Checking for valid age



The screenshot shows the Eclipse IDE with the `ValidationUtils.java` file open. The code defines a `ValidationUtils` class with two static methods: `validateAge(int age)` and `validateGrade(int grade)`. The `validateAge` method returns `true` if the age is greater than 0, and `validateGrade` returns `true` if the grade is between 0 and 100. The console output shows the execution of the `StudentInformationSystem` application, which prompts the user to enter their choice. The user enters 1, and the application prompts for student details. The user enters 105 for the student ID, om for the name, 0 for the age, and 87 for the grade. The application prompts for the contact number, and the user enters 6578566313. The application then adds the student successfully and displays the menu again.

```
package INTERNSHIP;

public class ValidationUtils {
    public static boolean validateAge(int age) {
        return age > 0;
    }

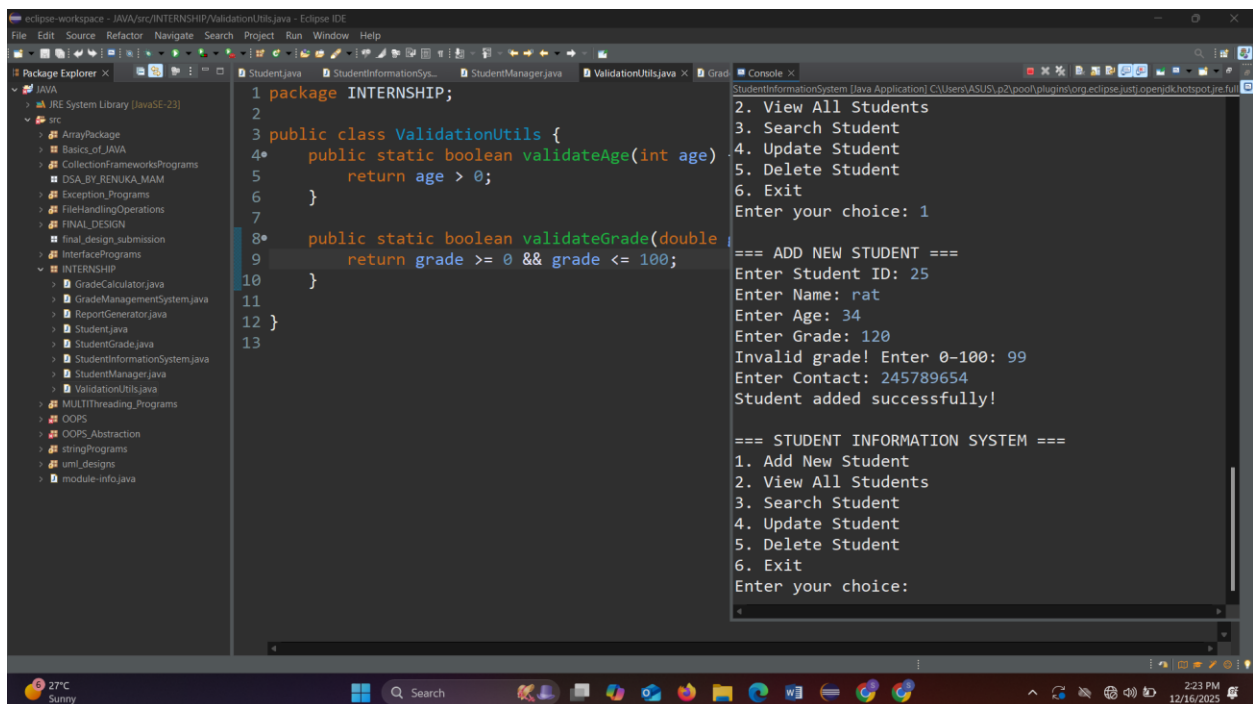
    public static boolean validateGrade(int grade) {
        return grade >= 0 && grade <= 100;
    }
}

StudentInformationSystem [Java Application] C:\Users\ASUS\AppData\Local\Temp\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.23.0\jre\bin\java.exe
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 1

=== ADD NEW STUDENT ===
Enter Student ID: 105
Enter Name: om
Enter Age: 0
Invalid age! Enter again:
22
Enter Grade: 87
Enter Contact: 6578566313
Student added successfully!

=== STUDENT INFORMATION SYSTEM ===
1. Add New Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice:
```

Checking for valid grades



The screenshot shows the Eclipse IDE with the `ValidationUtils.java` file open. The code defines a `ValidationUtils` class with two static methods: `validateAge(int age)` and `validateGrade(double grade)`. The `validateAge` method returns `true` if the age is greater than 0, and `validateGrade` returns `true` if the grade is between 0 and 100. The console output shows the execution of the `StudentInformationSystem` application, which prompts the user to enter their choice. The user enters 1, and the application prompts for student details. The user enters 25 for the student ID, rat for the name, 34 for the age, and 120 for the grade. The application prompts for the contact number, and the user enters 245789654. The application then adds the student successfully and displays the menu again.

```
package INTERNSHIP;

public class ValidationUtils {
    public static boolean validateAge(int age) {
        return age > 0;
    }

    public static boolean validateGrade(double grade) {
        return grade >= 0 && grade <= 100;
    }
}

StudentInformationSystem [Java Application] C:\Users\ASUS\AppData\Local\Temp\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.23.0\jre\bin\java.exe
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 1

=== ADD NEW STUDENT ===
Enter Student ID: 25
Enter Name: rat
Enter Age: 34
Enter Grade: 120
Invalid grade! Enter 0-100: 99
Enter Contact: 245789654
Student added successfully!

=== STUDENT INFORMATION SYSTEM ===
1. Add New Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice:
```

6. HOW THE PROJECT MEETS TECHNICAL REQUIREMENTS

The Student class represents the core data model and stores essential attributes such as student ID, name, age, grade, and contact information, following encapsulation principles with private variables and public getters and setters.

Data validation is handled through the ValidationUtils class, which ensures that the student's age is a positive number and the grade value lies between 0 and 100. This prevents invalid data from being entered into the system and improves data reliability.

The application provides a menu-driven console interface using the Scanner class in StudentInformationSystem. A switch-case structure allows users to interact efficiently with the system by selecting options to add, view, search, update, or delete student records.

CRUD operations are implemented through the StudentManager class. It allows creating new student records, reading and displaying all students in a formatted output, updating existing student details, and deleting student records dynamically using an ArrayList.

Search functionality is implemented to locate student records by either student ID or name, making record retrieval fast and user-friendly. Student data is stored and managed during runtime using an ArrayList, which allows flexible data handling such as dynamic addition and removal of records.

The output is displayed in a readable and aligned format using simple System.out.println() statements with tab spacing, ensuring clarity without using advanced formatting functions. Basic error handling is also included to manage invalid menu choices and incorrect input types, preventing the application from crashing.

7. CONCLUSION

This project demonstrates understanding of arrays, loops, OOP, input handling, and data processing in Java.